Web Based Diabetic Retinopathy Detection using Deep Learning

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INTRODUCTION

Problem Statement:

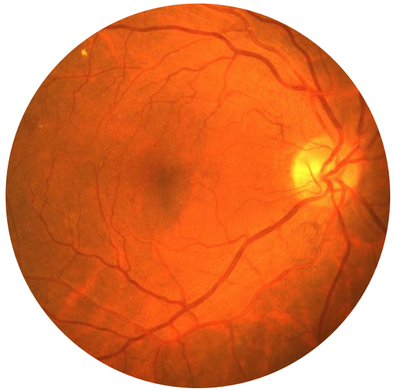
- ***Identification of Diabetic Retinopathic condition using Retinal Fundus images***
- **To develop and deploy a ML model**
- ***Results in one stop solution for identification, classification and deployment of DR model using Deep Learning Techniques.***

DATASET

APTOS Dataset:



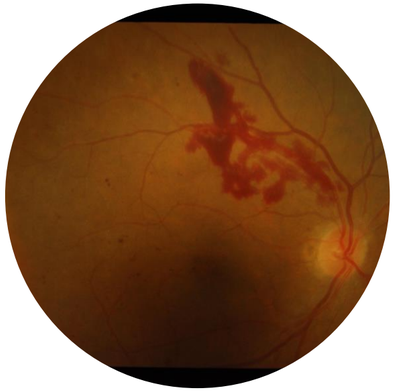
No DR



Mild DR



Moderate DR



Severe DR

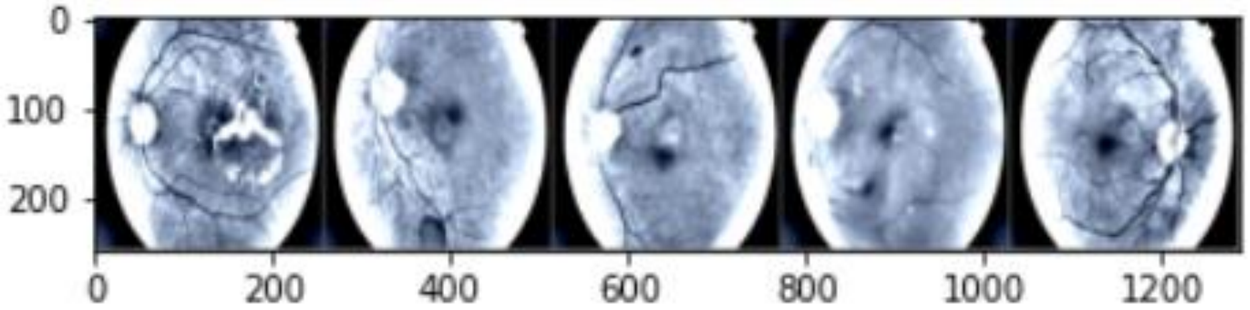


Proliferate DR

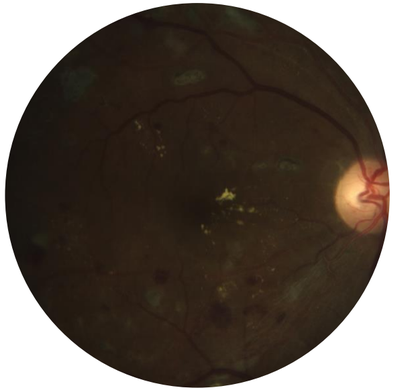
Data Preprocessing:

Image Resizing and cropping

Gray-Scaling



Balancing



Dropping noisy images

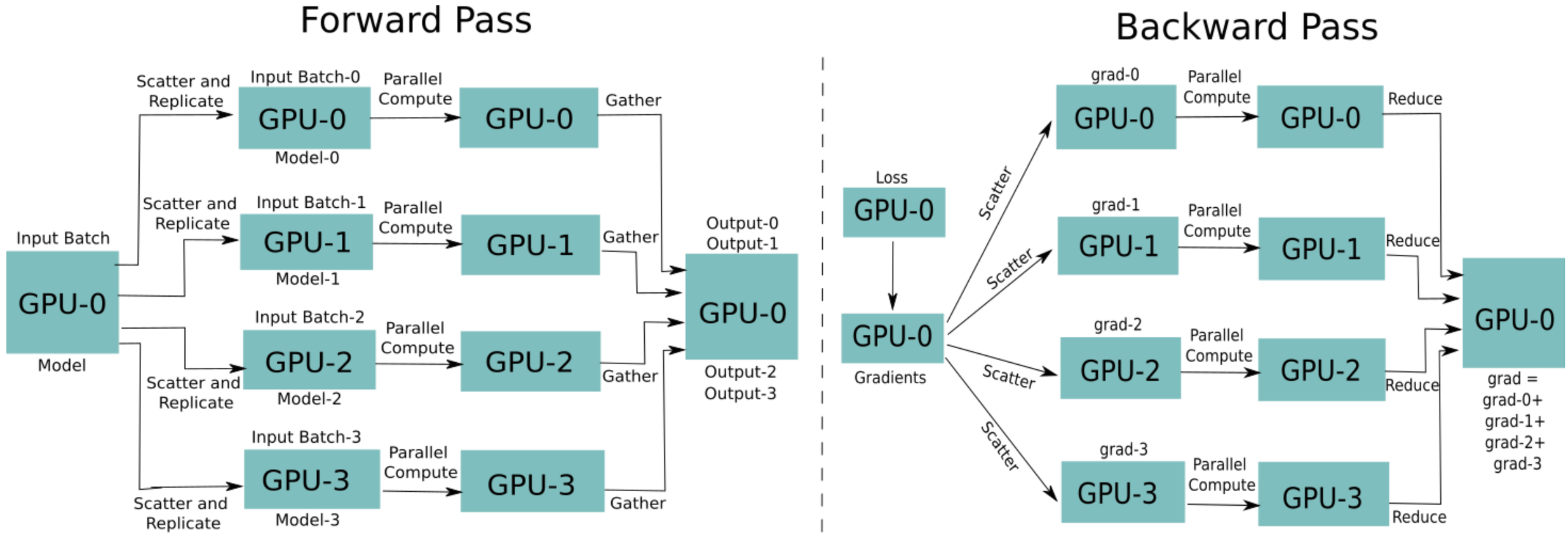
DEEP LEARNING MODELS

- Vision Transformer
- ResNet
- DenseNet

Implementation:

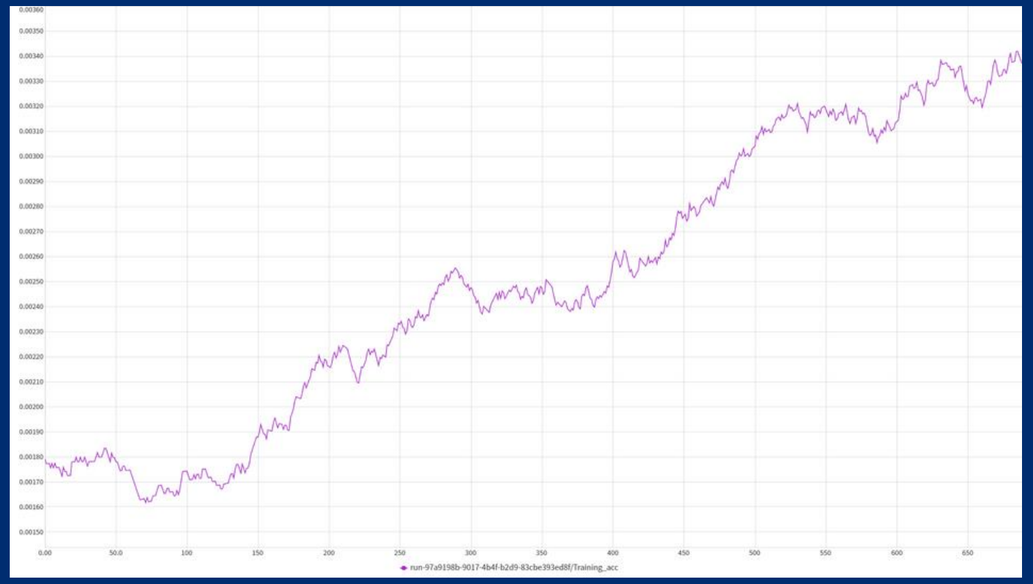
- Personal Computer
- High Performance Computer

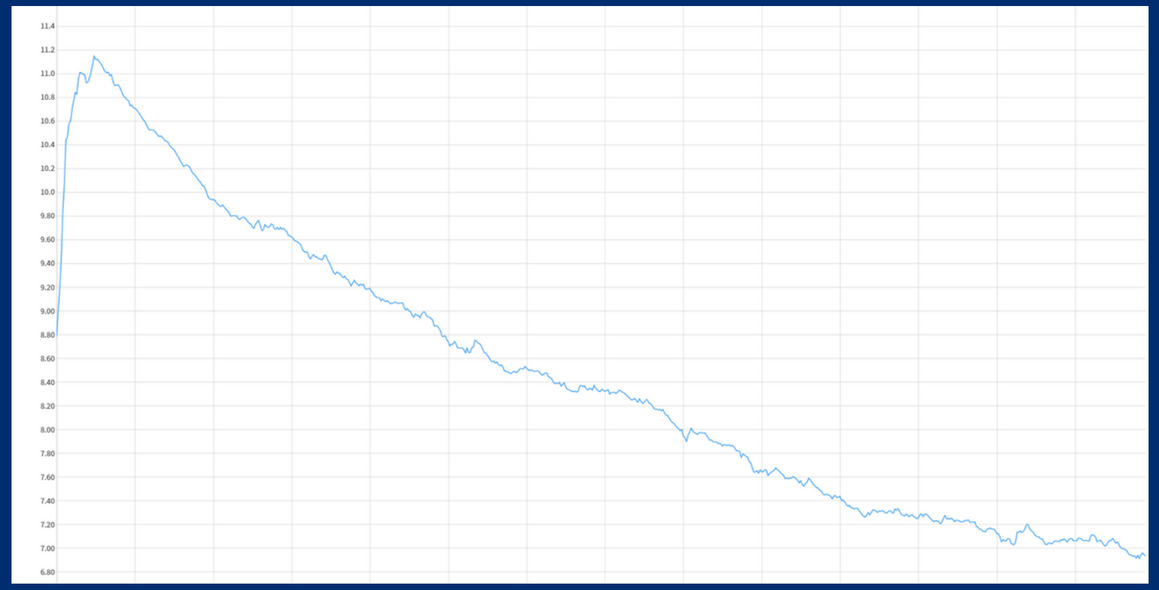
HPC ARCHITECTURE



Supporting mechanisms used: CUDA, NCCL, DDP

MODEL EVALUATION: ViT

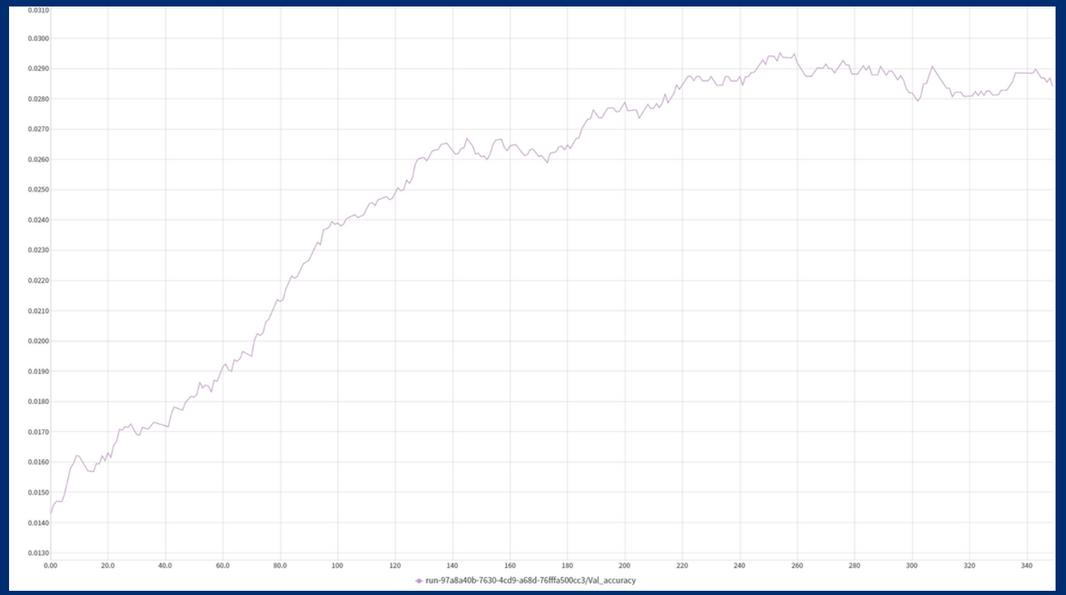




Training accuracy

Training loss

validation accuracy



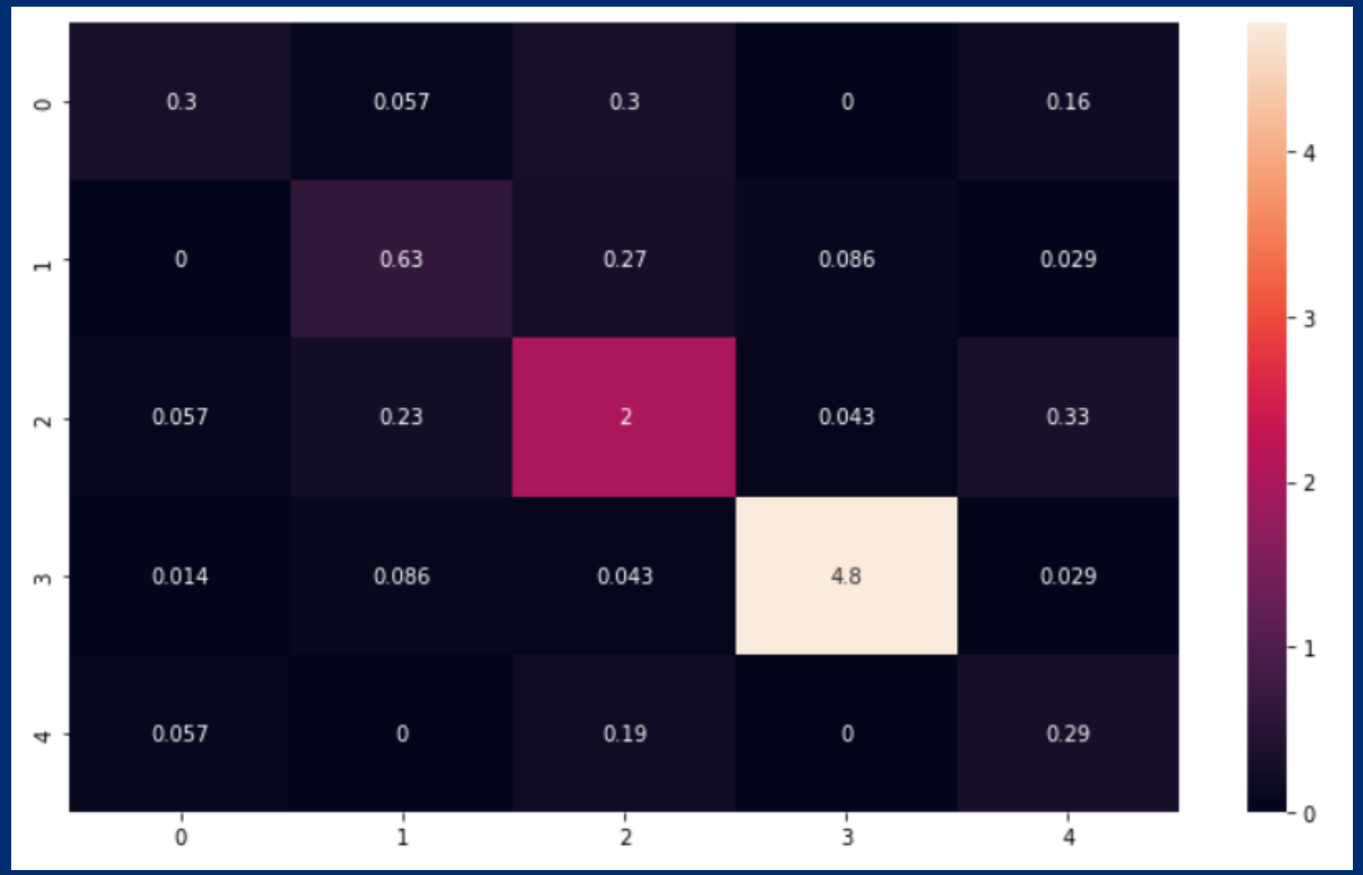
MODEL EVALUATION: RESNET

01 **ACCURACY:** 0.814072

02 **PRECISION:** 0.672866

02 **RECALL :** 0.649863

02 **F1 SCORE :** 0.646639



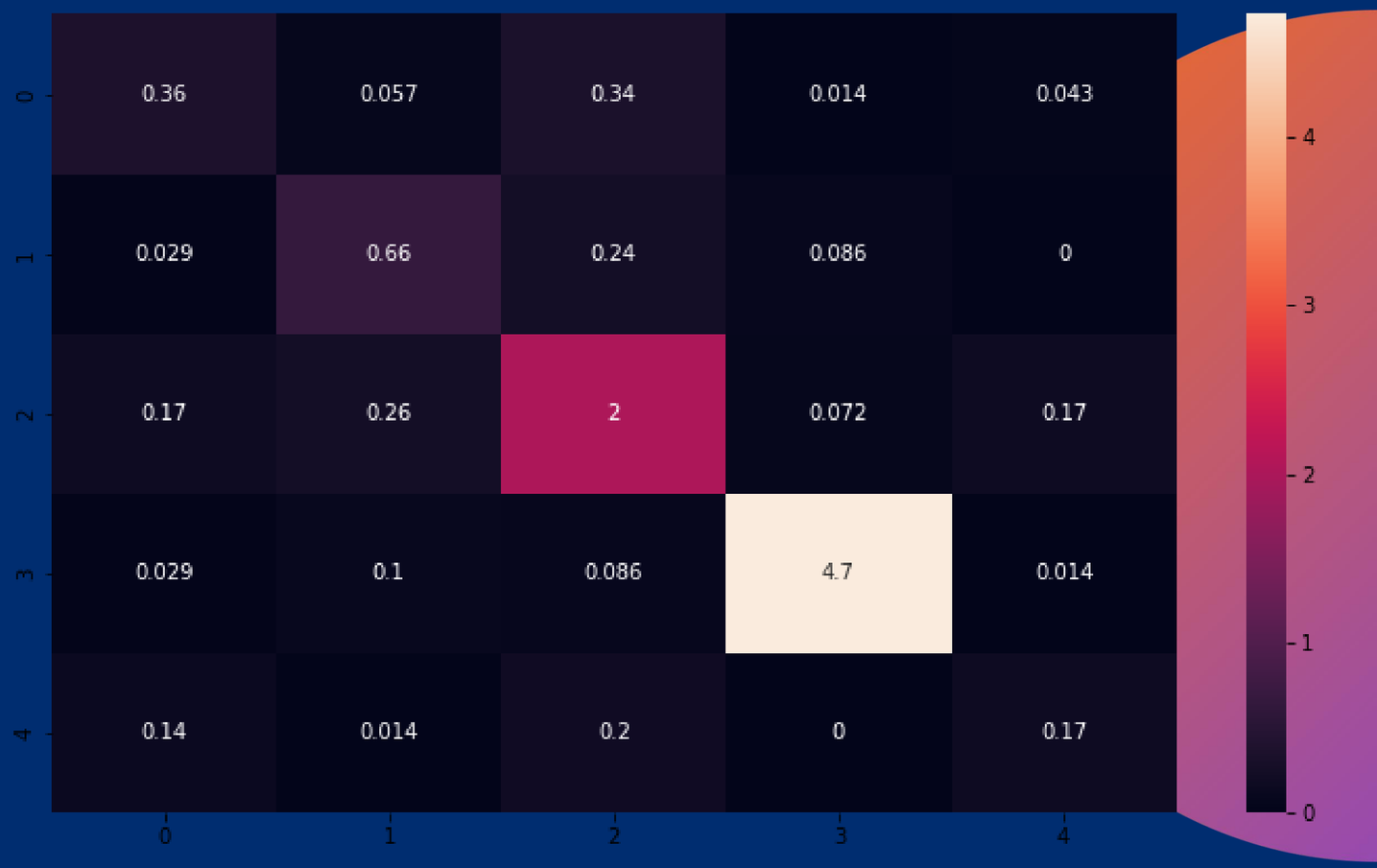
MODEL EVALUATION: DENSENET

01 **ACCURACY:** 0.78949

02 **PRECISION:** 0.63739

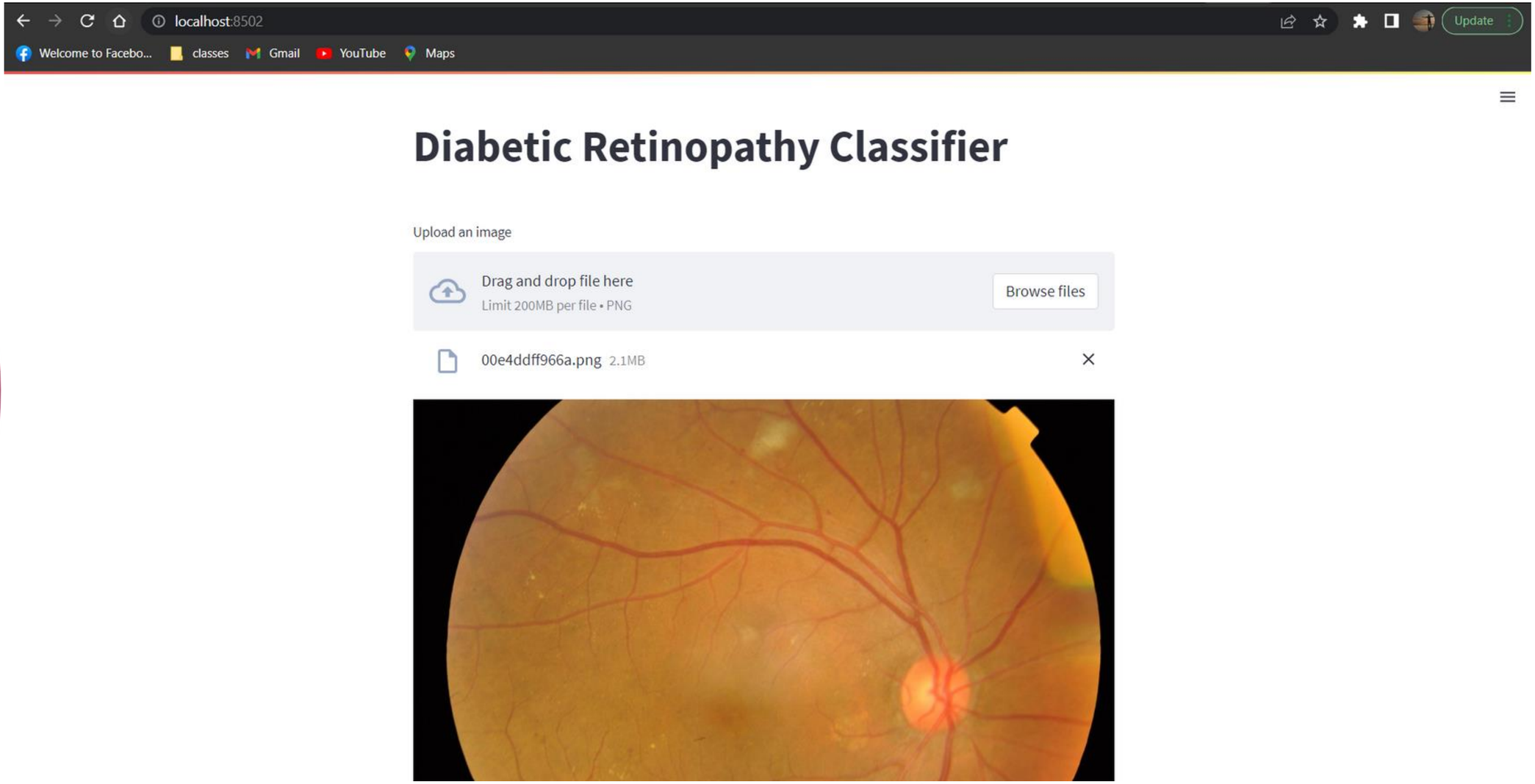
02 **RECALL :** 0.62291

02 **F1 SCORE :** 0.62808



MODEL DEPLOYMENT

APPLICATION USING STREAM-LIT



COMPARISON

01

MODEL PERFORMANCE

- 81 % and 79 % for Resnet and Densenet respectively

02

TIME TAKEN

- ViT took most time while Resnet scored least

03

MODEL COMPLEXITY

- ViT is the most complex model with different layers

LIMITATIONS

- Highly imbalanced dataset
- Noisy images
- Limited availability of HPC clusters
- Limited free cloud resources

FUTURE WORK

- Better training with HPC clusters
- Model deployment with CI/CD methodologies
- Advancement in de- centralised training
- Development of light-weight models

References

https://naga-karthik.github.io/post/pytorch-ddp/